



E-WASTE TRACKING AND DECISION-SUPPORT SYSTEM REPORT

Green Computing Midterm Project

Prepared by

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1. Abstract

Electronic waste is a rapidly growing sustainability challenge because electronic devices contain both recoverable materials and hazardous substances. Universities continuously acquire, use, relocate, repair, and retire computers, displays, batteries, networking devices, and laboratory equipment. Without a structured inventory and decision process, obsolete equipment may remain in storage, lose reuse value, be mixed with general waste, or enter informal recycling channels. This report develops a report-centered e-waste tracking and decision-support framework for East West University. A simulated dataset of 100 items is used to demonstrate the required data fields, classification logic, dashboard indicators, and environmental impact calculations. The methodology applies transparent rules based on condition, functionality, safety, device age, repair cost, replacement cost, and recyclability. The proposed outputs are reuse, repair, recycle, and safe disposal. The simulated evaluation indicates that 90% of items can be directed to circular pathways before controlled disposal, while a weight-based diversion rate of approximately 89.0% is achieved. The report also discusses feasibility, scalability, privacy, maintenance, stakeholder responsibilities, and limitations. The proposed framework is not a production system; it is a validated report-level model that can guide a later pilot implementation.

2. Introduction

2.1 Background

Electronic waste, commonly called e-waste, includes discarded electrical and electronic equipment such as computers, monitors, mobile phones, batteries, printers, networking devices, and accessories. E-waste is environmentally significant because it combines valuable resources - including metals, plastics, and reusable components - with substances that can be harmful when devices are dismantled, burned, dumped, or stored without proper safeguards.

The World Health Organization reports that 62 million tonnes of e-waste were generated globally in 2022 and only 22.3% was documented as formally collected and recycled. The same source identifies e-waste as one of the fastest-growing solid-waste streams and warns that informal handling can release hazardous substances such as lead. These global findings justify the need for stronger institutional tracking and responsible routing of end-of-life electronics [\[1\]](#) [\[2\]](#) .

Universities are technology-intensive institutions. Academic laboratories, administrative offices, libraries, faculty workspaces, and student-service units rely on devices with different purchase dates, utilization levels, maintenance histories, and security requirements. As a result, a university may simultaneously hold usable surplus equipment, repairable equipment, obsolete but recyclable equipment, and hazardous devices requiring controlled disposal.

2.2 Green Information Systems Perspective

Green Information Systems focuses on using information, analytics, governance, and digital processes to reduce environmental impacts and improve resource efficiency. In this project, the value of computing is not limited to building software. The principal contribution is a structured information process that makes e-waste visible, classifies it consistently, estimates environmental benefits, and supports accountable decisions.

2.3 Significance of the Study

- Reduces unnecessary disposal by identifying reusable and repairable equipment.
- Improves resource recovery by separating recyclable devices from general waste.
- Supports safer handling of batteries, damaged displays, and other potentially hazardous components.
- Creates auditable evidence for institutional sustainability reporting and management decisions.
- Provides a foundation for future integration with procurement, asset management, and approved recycler records.

3. Problem Statement

The selected sustainability problem is the absence of a consistent, reportable, and decision-oriented process for managing institutional e-waste. Individual departments may know that equipment is old or damaged, but this knowledge is often fragmented across spreadsheets, maintenance notes, storage rooms, and personal communication. A device may be marked simply as "unused" even though the correct decision could be reassignment, component repair, recycling, data destruction, or controlled disposal.

This fragmentation produces five core risks. First, useful devices may be discarded prematurely. Second, repairable devices may remain idle until their condition deteriorates. Third, storage rooms may accumulate undocumented equipment. Fourth, storage media may leave the institution without confirmed data sanitization. Fifth, hazardous or non-functional devices may enter inappropriate disposal channels.

3.1 System Boundary

The proposed boundary starts when a department identifies an electronic item as surplus, damaged, obsolete, or no longer required. It includes data registration, inspection, classification, decision recommendation, approval, handover, environmental-impact estimation, and status reporting. It ends when the item is reassigned, repaired and returned, transferred to an approved recycling channel, or documented as safely disposed of. Procurement planning and vendor contracting are discussed as interfaces but are not modeled as full modules.

3.2 Stakeholders

Stakeholder	Primary interest	Role in the proposed process
IT Services	Asset condition, data security, technical feasibility	Inspection, data sanitization, technical recommendation
Department/Asset Custodian	Operational continuity and accountability	Reports surplus or damaged items and confirms handover
Administration/Procurement	Cost control and policy compliance	Approves repair, replacement, transfer, or disposal
Finance/Internal Audit	Traceability and asset write-off evidence	Verifies records and financial treatment
Facilities/Store	Safe temporary storage	Maintains labeled, segregated storage
Approved Repairer/Recycling Partner	Service and material recovery	Provides quotations, service evidence, and certificates
Sustainability Committee	Environmental performance	Reviews diversion, reuse, recycling, and improvement indicators

4. Assumptions and Data Source

4.1 Data Source

The analysis uses a purpose-built simulated dataset containing 100 e-waste records. Simulation is appropriate for this midterm project because no authorized East West University asset register was supplied. The dataset is designed to be plausible for a university environment but must not be interpreted as an actual inventory of the institution.

4.2 Main Assumptions

Assumption area	Scenario assumption	Reason for use
Inventory size	100 items across ten categories	Provides enough variation for tables and charts without implying a full institutional audit
Condition assessment	Good, Fair, Minor Fault, Major Fault, Obsolete, Non-functional, or Unsafe	Supports transparent decision rules
Repair threshold	Repair is normally recommended when estimated repair cost is 40% or less of replacement cost	Simple economic screening rule; subject to policy validation
Reuse eligibility	Functional, safe, and suitable for another user or department	Prioritizes life extension before recycling
Recycling eligibility	Non-economical to repair but recoverable through an approved channel	Supports material recovery and documented handover
Safe disposal	Leaking, hazardous, contaminated, or otherwise unsafe equipment	Prevents routine handling or informal disposal
Impact factors	Reuse 3.0, repair 2.2, recycle 1.1, safe handling 0.15 kg CO ₂ e benefit per kg	Illustrative scenario coefficients, not audited LCA factors
Financial value	Avoided purchase, repair-versus-replacement difference, or conservative scrap recovery value	Produces decision-support estimates only

4.3 Data Fields

Field	Type	Purpose
Item ID	Text	Unique traceability
Category	Categorical	Device grouping and reporting
Department	Categorical	Custodian and location accountability
Age	Numeric	Obsolescence and lifecycle context
Condition	Categorical	Physical and functional assessment
Weight	Numeric	Waste and environmental calculations
Repair cost	Currency	Repair feasibility
Replacement cost	Currency	Economic comparison
Hazardous component	Yes/No	Safety and disposal routing
Recommended action	Categorical	Reuse, repair, recycle, or safe disposal
Decision rationale	Text	Explainability and approval support
Final status/date	Status/date	Closure and audit evidence
Handover certificate	Reference	Evidence from repairer or recycler

5. Proposed System Design

The design is intentionally modular so that the institution can begin with a spreadsheet or notebook and later migrate to a database-backed application. The report separates data capture, validation, decision logic, impact estimation, and reporting. This separation reduces the risk that a visually attractive dashboard hides weak or inconsistent source data.

Proposed Report-Level Architecture

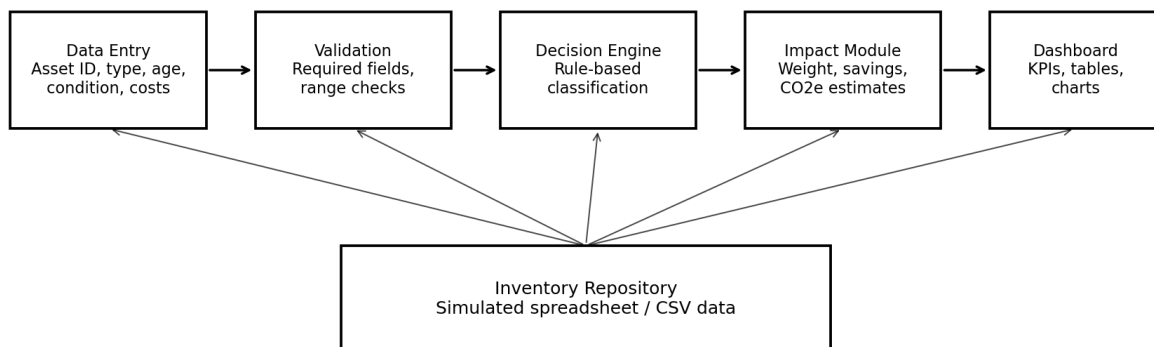


Figure 1. Proposed report-level architecture for e-waste tracking and decision support.

5.1 Architecture Components

Component	Function
Data-entry layer	Captures item identity, department, age, condition, safety indicators, costs, and weight.
Validation layer	Checks mandatory fields, duplicate IDs, impossible values, missing costs, and inconsistent status combinations.
Inventory repository	Stores the structured records. For the project, this is a CSV/spreadsheet dataset.
Decision engine	Applies the documented rule hierarchy and returns one recommended pathway with a rationale.
Impact module	Calculates item counts, weight, reuse/recycling rates, direct savings, and illustrative CO2e benefits.
Reporting/dashboard layer	Displays KPIs, category distribution, action distribution, unresolved items, and handover status.

5.2 Data Flow

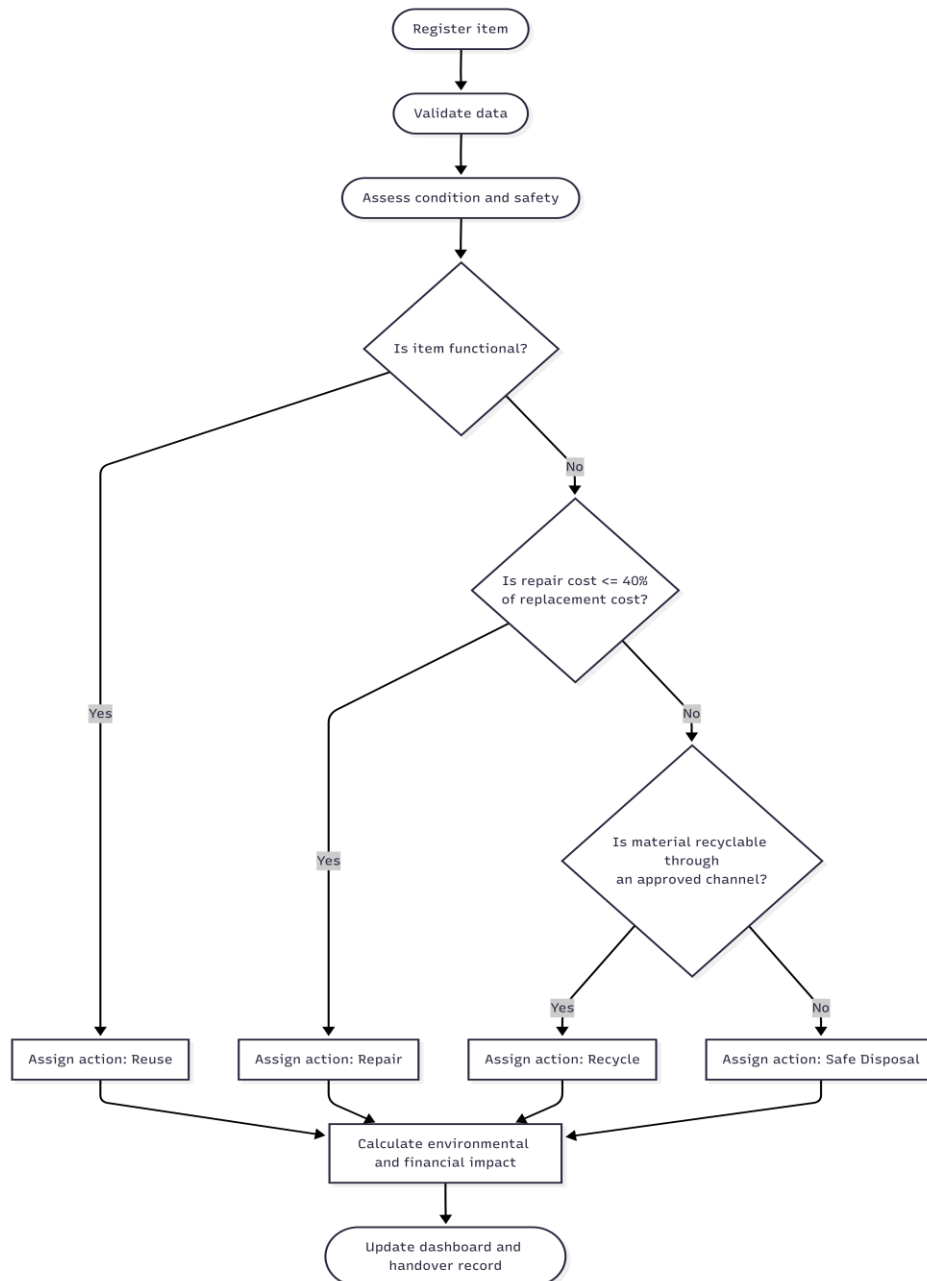


Figure 2. Decision and reporting flow from item registration to documented outcome.

5.3 Suggested Tools

For the current project, Microsoft Excel or Google Sheets can be used for item entry and formulas, while Python with pandas and Matplotlib can generate reproducible calculations and figures. A later pilot may use a lightweight web dashboard such as Streamlit or Power BI. A full web system is not necessary to demonstrate the academic contribution of the report.

6. Methodology

6.1 Condition Classification

Class	Operational definition	Typical evidence
Good	Fully functional, safe, and suitable for continued use	Passes functional test; no significant damage
Fair	Functional but cosmetically worn or lower specification	Minor wear; still supports a valid use case
Minor Fault	One or more repairable faults without major safety risk	Replaceable keyboard, fan, RAM, drive, cable, or power unit
Major Fault	Multiple or expensive faults	Main board, panel, structural, or repeated power failure
Obsolete	Functional or partly functional but no longer secure or supportable	Unsupported operating system, incompatible interface, inadequate capability
Non-functional	Does not operate and repair is not justified	Fails power-on or essential functional tests
Unsafe/Hazardous	Leaking, swollen, broken, contaminated, or electrically unsafe	Battery swelling, leakage, broken display, burn damage

6.2 Decision Rules

The rule hierarchy gives priority to safety, then life extension, then material recovery. A lower-priority economic outcome cannot override a higher-priority safety requirement.

1. If an item is leaking, swollen, contaminated, severely broken, or otherwise unsafe, classify it for safe disposal or specialist hazardous handling.
2. Otherwise, if the item is functional, secure, and suitable for another institutional use, recommend reuse or reassignment.
3. Otherwise, calculate the repair ratio: repair cost divided by replacement cost.
4. If the item is repairable, parts are available, and the repair ratio is at most 0.40, recommend repair.
5. If repair is not economical but the item contains recoverable material and an approved recycling channel exists, recommend recycling.
6. If no safe reuse, repair, or recycling pathway is available, recommend controlled disposal with documented authorization.

6.3 Equations

Let N be the total number of assessed items, n_r the number reused, n_p the number repaired, n_c the number recycled, and n_d the number safely disposed. Let W be total item weight and W_d the weight assigned to safe disposal.

$$\text{Reuse Rate (\%)} = (n_r / N) * 100$$

$$\text{Repair Rate (\%)} = (n_p / N) * 100$$

$$\text{Recycling Rate (\%)} = (n_c / N) * 100$$

$$\text{Circular-pathway Rate (\%)} = ((n_r + n_p + n_c) / N) * 100$$

$$\text{Weight Diversion Rate (\%)} = ((W - W_d) / W) * 100$$

$$\text{Repair Ratio} = \text{Estimated Repair Cost} / \text{Estimated Replacement Cost}$$

$$\text{Estimated CO}_2\text{e Benefit} = \text{Sum of (Item Weight} \times \text{Action-specific Scenario Factor)}$$

$$\text{Estimated Direct Saving} = \text{Avoided Purchase Value} + \text{Repair Saving} + \text{Material Recovery Value}$$

The environmental coefficients used in this report are illustrative. A real institutional claim should use locally relevant life-cycle inventory data or documented recycler reports. The purpose here is to demonstrate the calculation framework, not to certify a carbon footprint.

6.4 Algorithm

The following pseudocode summarizes the decision process:

```

FOR each item:
  validate mandatory fields
  IF unsafe or hazardous leakage: action = SAFE DISPOSAL
  ELSE IF functional and suitable: action = REUSE
  ELSE compute repair_ratio
    IF repairable AND repair_ratio <= 0.40: action = REPAIR
    ELSE IF approved recycling route exists: action = RECYCLE
    ELSE: action = SAFE DISPOSAL
  calculate weight, financial, and environmental indicators
  store decision rationale and approval status

```


7. Computational Artifact

The computational artifact consists of three elements: (1) the simulated CSV inventory, (2) a reproducible rule-and-calculation workflow, and (3) dashboard-style figures embedded in this report. This design satisfies the requirement to demonstrate a computational result while keeping the main focus on analysis and reporting.

7.1 Artifact Inputs

- One row per electronic item.
- Standardized categories and condition labels.
- Numeric weight, repair cost, and replacement cost.
- Safety indicator and decision rationale.
- Action-specific environmental scenario factors.

7.2 Artifact Outputs

- Item-level recommended action.
- Count and weight by device category.
- Count and weight by management action.
- Reuse, repair, recycling, and circular-pathway rates.
- Estimated CO2e benefit and direct financial value.
- Visual summaries for management discussion.

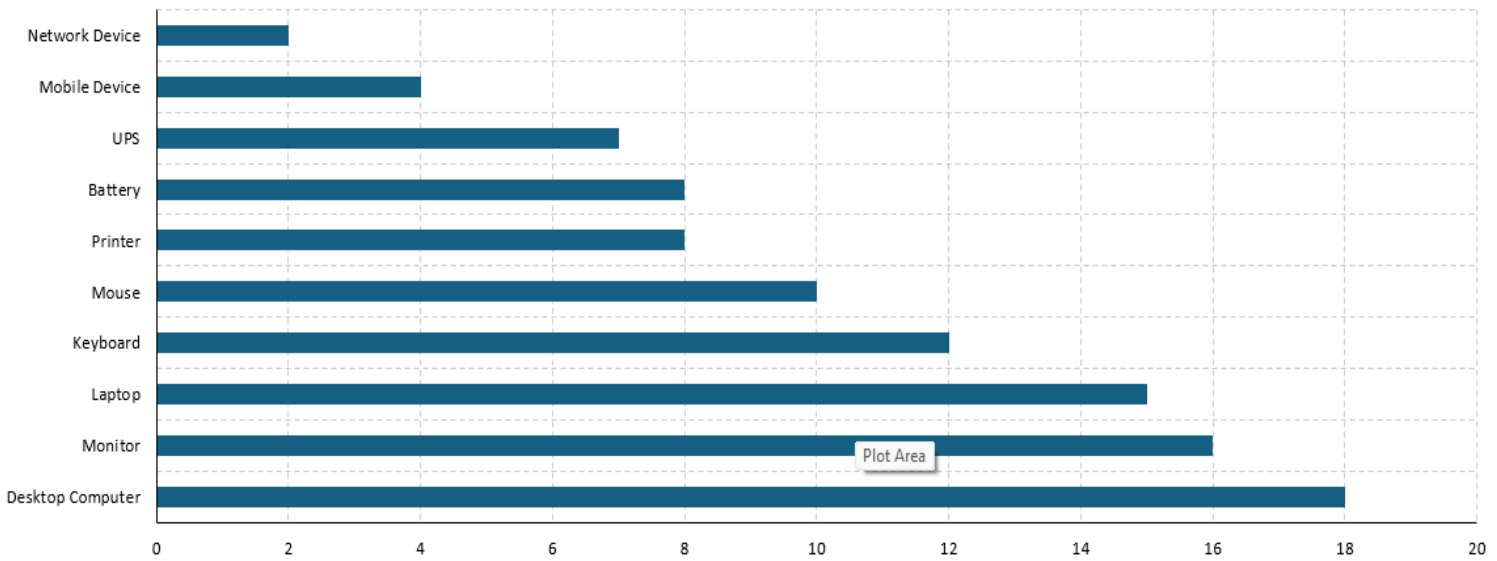


Figure 3. Distribution of the 100 simulated items by device category.

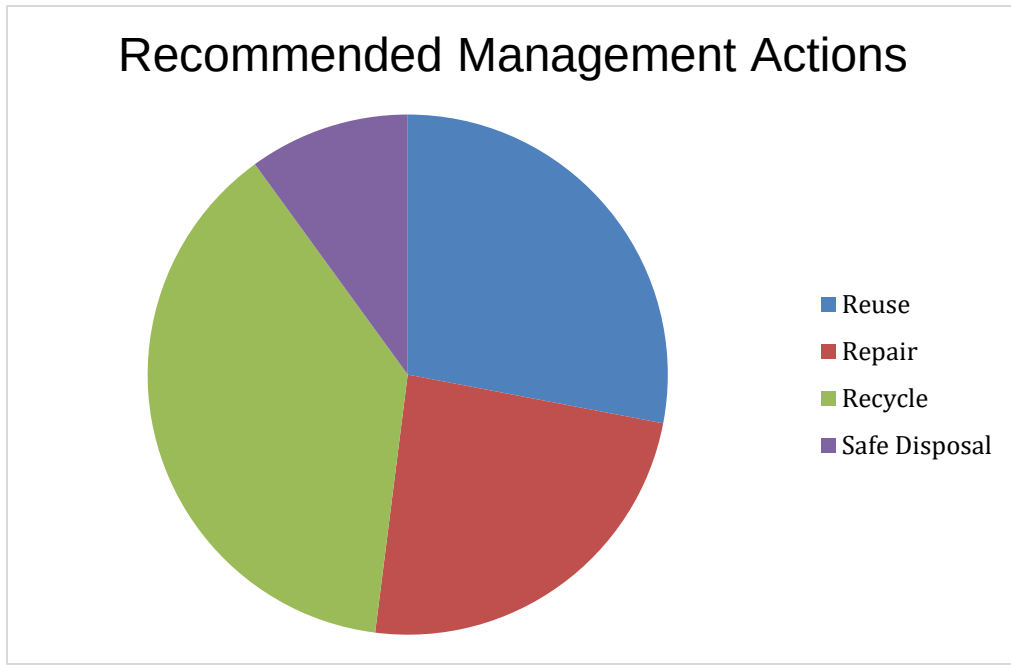


Figure 4. Recommended actions produced by the rule-based framework.

8. Results and Evaluation

8.1 Inventory Summary

Category	Items	Estimated weight (kg)
Desktop Computer	18	158.7
Monitor	16	86.8
Laptop	15	32.3
Keyboard	12	8.2
Mouse	10	1.6
Battery	8	20.1
Printer	8	89.3
UPS	7	54.5
Mobile Device	4	0.8
Network Device	2	3.2

The simulated inventory contains 100 items with a combined estimated weight of 455.5 kg. Desktop computers, monitors, and laptops account for the largest item counts, while printers, desktop computers, monitors, and UPS units contribute a large share of total mass. This difference shows why both item count and weight should be reported; a large number of low-mass peripherals can otherwise distort management priorities.

8.2 Decision Outcomes

Action	Items	Share	Weight kg	CO2e benefit kg	Estimated value BDT
Reuse	28	28.0%	102.9	308.6	542,918
Repair	24	24.0%	132.6	291.7	592,500
Recycle	38	38.0%	169.9	186.9	20,376
Safe Disposal	10	10.0%	50.1	7.5	0

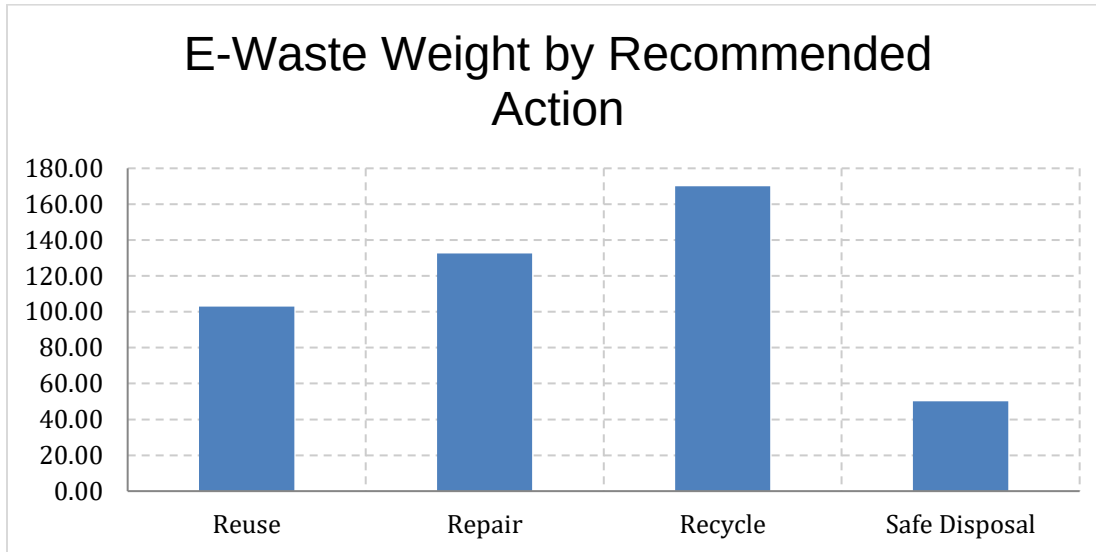


Figure 5. Estimated e-waste mass associated with each recommended action.

8.3 Key Performance Indicators

Indicator	Result	Interpretation
Total assessed items	100	Size of simulated case-study inventory
Total estimated weight	455.5 kg	Mass requiring accountable management
Reuse rate	28.0%	Items suitable for reassignment or continued use
Repair rate	24.0%	Items where repair is economically preferred
Recycling rate	38.0%	Items routed to material recovery
Circular-pathway rate	90.0%	Items assigned to reuse, repair, or recycling
Weight diversion rate	89.0%	Mass kept away from uncontrolled disposal in the scenario
Estimated CO2e benefit	794.8 kg CO2e	Illustrative scenario estimate
Estimated direct value	BDT 1,155,794	Avoided purchase, repair saving, and recovery value

8.4 Before-and-After Comparison

Dimension	Without structured process	With proposed framework
Visibility	Items distributed across rooms and informal lists	One standardized register with unique IDs
Decision consistency	Depends on individual judgment	Documented safety and economic rules
Reuse opportunity	Often discovered late or not at all	Reuse considered before repair or recycling
Data security	Storage media handling may be undocumented	Sanitization checkpoint and custody record
Environmental reporting	No reliable count, weight, or diversion indicators	Repeatable KPIs and visual summaries
Accountability	Unclear ownership and handover status	Custodian, approver, date, and certificate fields
Management value	Reactive disposal decisions	Evidence for budgeting, repair planning, and recycler selection

9. Deployment Considerations

9.1 Feasibility

A first-stage implementation is technically feasible because it can use common office tools. The institution can begin with a protected spreadsheet and a written inspection checklist. The main requirement is governance: departments must use the same fields, condition labels, approval process, and handover evidence.

10.2 Scalability

The structure can scale from a single laboratory to the entire university. At larger scale, role-based access, barcode or QR labels, bulk imports, approval workflows, and integration with fixed-asset records become more important. The data model should remain stable even if the interface changes.

10.3 Privacy and Data Security

E-waste records may reveal device serial numbers, office locations, custodians, maintenance history, and storage media status. Access should therefore follow least-privilege principles. Storage devices should be sanitized using an approved method before reuse, external repair, recycling, or disposal. The sanitization result and responsible person should be recorded.

10.4 Cost

The spreadsheet/notebook version has low software cost but requires staff time for inspection, labeling, data entry, and review. A later system would add development, hosting, training, support, backup, and integration costs. These costs should be compared with avoided replacement purchases, reduced storage, improved repair planning, and lower compliance risk.

10.5 Maintenance and Data Quality

Data quality is the largest operational risk. Required fields, controlled lists, duplicate checking, periodic audits, and closure evidence are more important than advanced visual design. Records should not remain indefinitely in a "pending" state. A monthly or quarterly review should identify overdue inspection, repair, recycling, or certificate collection tasks.

10.6 Legal and Environmental Alignment

Bangladesh issued the Hazardous Waste (E-waste) Management Rules, 2021, and the Department of Environment lists e-waste rules among the country's environmental rules. A real university process should be reviewed against current requirements and use authorized service providers where applicable [\[4\]](#).

Internationally, the Basel Convention provides technical guidance concerning e-waste and the distinction between waste and non-waste in transboundary movement [\[3\]](#).

10.7 Deployment Risk Register

Risk	Likelihood	Impact	Mitigation
Incomplete or inaccurate data entry	High	High	Mandatory fields, training, and periodic sampling audit
Departments retain items without reporting	Medium	High	Policy deadline and custodian accountability
Repair estimates are unreliable	Medium	Medium	Two quotations above a defined value threshold
Informal recycler is selected	Medium	High	Approved-vendor list and handover certificate
Storage media leaves without sanitization	Low/Medium	Very High	Mandatory IT clearance and evidence field
Hazardous items stored with general equipment	Medium	High	Segregated labeled storage and rapid escalation
Carbon estimates are presented as audited facts	Medium	Medium	Label factors as scenario assumptions and document sources
Dashboard is updated but physical status is not	Medium	High	Periodic reconciliation and closure evidence

11. Stakeholder Responsibilities

Activity	Responsible	Accountable	Consulted/Informed
Identify surplus or damaged item	Department custodian	Department head	IT Services
Technical and safety inspection	IT Services	IT manager	Department custodian
Repair/replacement decision	IT/Procurement	Administration	Finance and user department
Data sanitization	Authorized IT personnel	IT manager	Internal audit where required
Temporary storage	Store/Facilities	Administration	IT and Safety personnel
Recycler or repairer selection	Procurement	Administration	IT, Finance, Sustainability Committee
Handover and certificate capture	Store/Procurement	Administration	Internal audit
KPI review and improvement	Sustainability Committee	Senior management	All participating departments

12. Limitations

- The dataset is simulated and does not describe the actual quantity or condition of e-waste at East West University.
- The environmental factors are illustrative scenario values, not locally verified life-cycle assessment coefficients.
- Repair and replacement costs are simulated and may differ substantially from market quotations and institutional contracts.
- The decision rules simplify technical diagnosis and cannot replace qualified inspection.
- The framework does not model every asset-accounting, procurement, or legal approval requirement.
- No real users were observed, and usability or inter-rater reliability of condition assessments was not tested.
- The report does not verify the availability or capacity of specific approved recyclers.

These limitations do not invalidate the framework; they define what must be validated during a real pilot. The most important next research step is to replace simulated records with an authorized sample audit and compare independent inspectors' classifications.

13. Conclusion and Future Improvement

This project demonstrates that a strong Green Information Systems contribution can be delivered through a rigorous report, transparent data structure, explainable rules, and measurable indicators without constructing a large production system. The proposed framework makes institutional e-waste visible and directs each item toward reuse, repair, recycling, or safe disposal. In the simulated 100-item case study, 90 items enter circular pathways and approximately 89.0% of total estimated mass is diverted from uncontrolled disposal. The scenario also demonstrates how direct financial value and environmental benefit can be summarized for management.

The recommended next step is a limited pilot covering one computer laboratory and one administrative unit. The pilot should validate field definitions, inspection consistency, repair thresholds, storage arrangements, data-sanitization evidence, and recycler documentation. After the process is stable, the institution may consider barcode labels, role-based workflows, integration with the asset register, automated reminders, and more rigorous life-cycle impact factors.

14. References

- [1] “E-waste and Children’s Health.” World Health Organization, 1 Oct. 2024, [www.who.int/news-room/fact-sheets/detail/electronic-waste-\(e-waste\)](http://www.who.int/news-room/fact-sheets/detail/electronic-waste-(e-waste)).
- [2] International Telecommunication Union and United Nations Institute for Training and Research, The Global E-waste Monitor 2024, 2024.
- [3] Basel Convention, "Technical guidelines on transboundary movements of electrical and electronic waste and used electrical and electronic equipment," adopted by COP-16, 2023.
- [4] Government of the People's Republic of Bangladesh, Hazardous Waste (E-waste) Management Rules, 2021, S.R.O. No. 187-Law/2021, 10 June 2021.